

Direct Torque Control of Induction Motor

DTC of IM

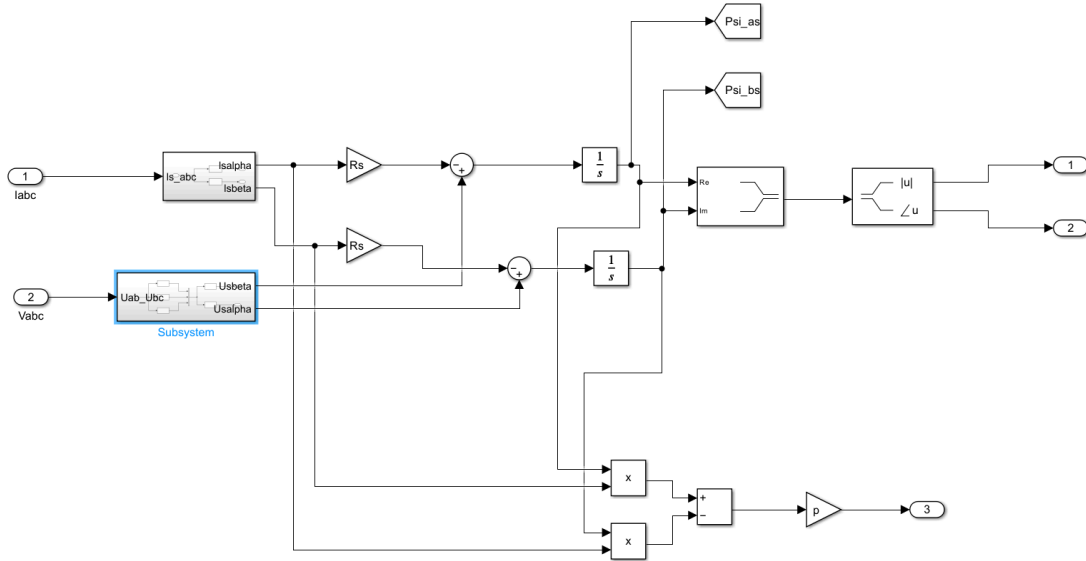


Figure 1.1: Stator flux, torque, and sector estimator subsystem used in the Direct Torque Control (DTC) model. The stator flux is calculated by integrating the difference between the stator voltage and resistive voltage drop, while the electromagnetic torque is found using the flux and current components. The magnitude and angle of the stator flux are also determined, which are used to decide the switching behavior of the controller.

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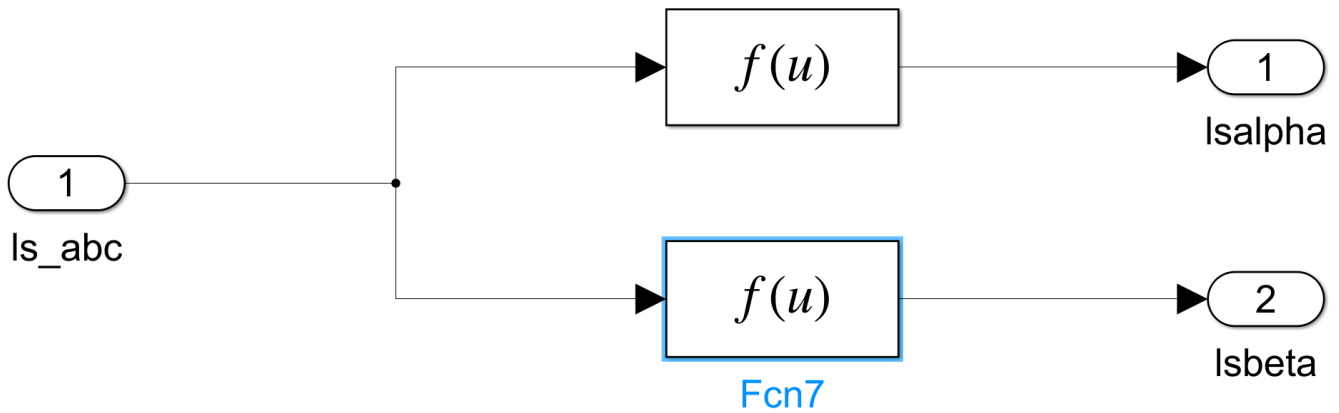


Figure 1.2: Iabc to Iαβ conversion subsystem. This block converts the three-phase stator currents into the αβ reference frame using a mathematical transformation. This makes the currents easier to use in the flux and torque calculations for the DTC controller.

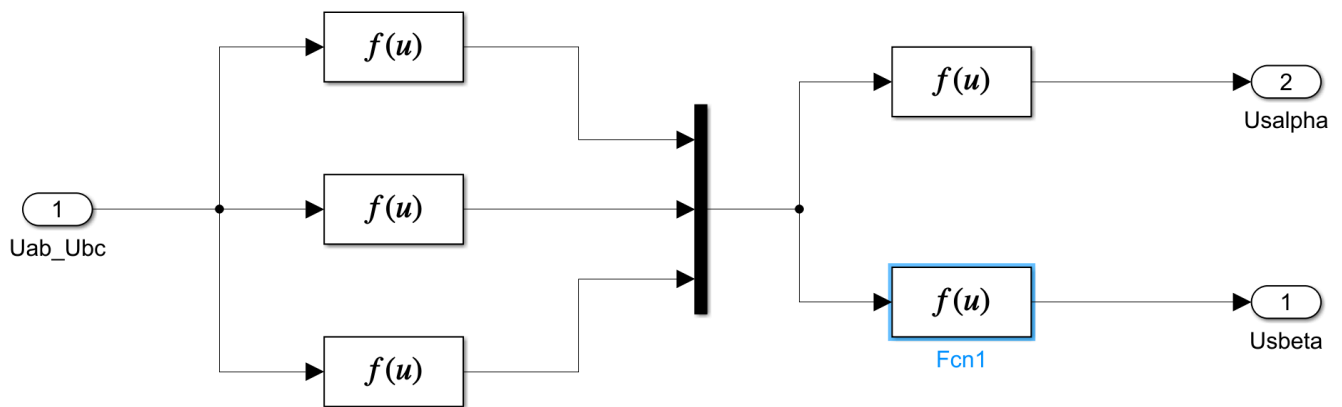


Figure 1.3: Uabc to Uαβ conversion subsystem. This block converts the three-phase input voltages into the αβ reference frame by first converting line voltages to phase voltages and then applying a transformation. These transformed voltages are used in calculating the stator flux for the DTC controller.

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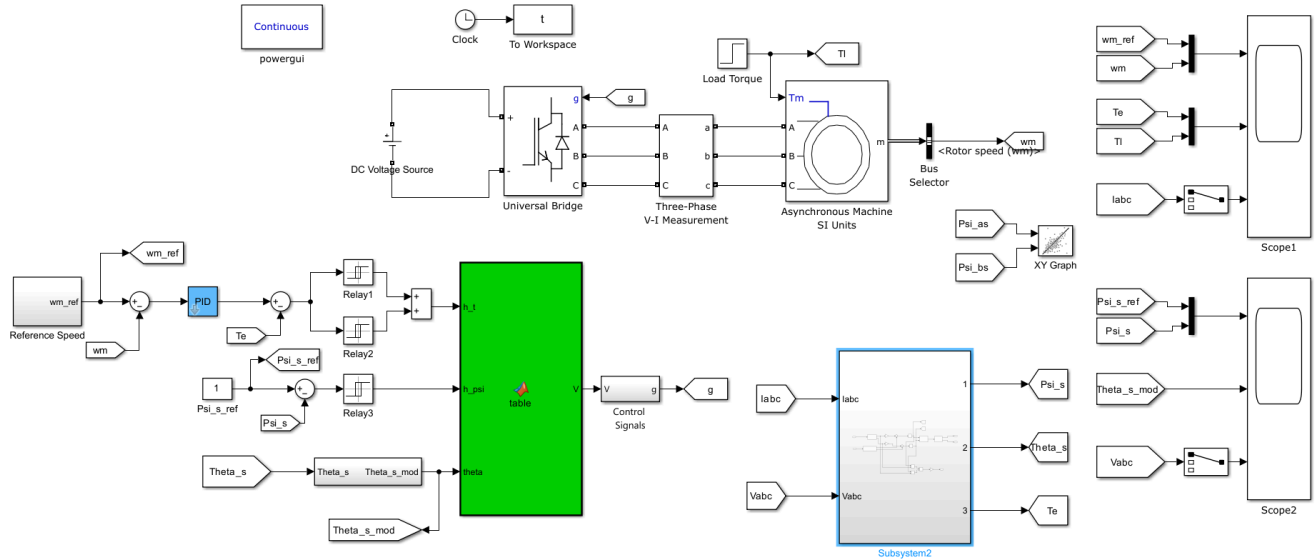


Figure 1.4: Complete Simulink model of the Direct Torque Control (DTC) system for an induction motor. The model includes the speed control loop, DTC switching logic, inverter, and induction motor, along with subsystems for current and voltage transformations and flux/torque estimation. This setup allows the controller to directly regulate torque and stator flux and monitor motor speed and performance.

$$\lambda_s = \sqrt{\left(\left(\int (V_\alpha - R_s i_\alpha) dt \right)^2 + \left(\int (V_\beta - R_s i_\beta) dt \right)^2 \right)}$$

$$\theta_s = \tan^{-1} \left(\frac{\lambda_\beta}{\lambda_\alpha} \right)$$

$$T_e = \frac{3}{2} P \cdot (\lambda_\alpha \cdot i_\beta - \lambda_\beta \cdot i_\alpha)$$

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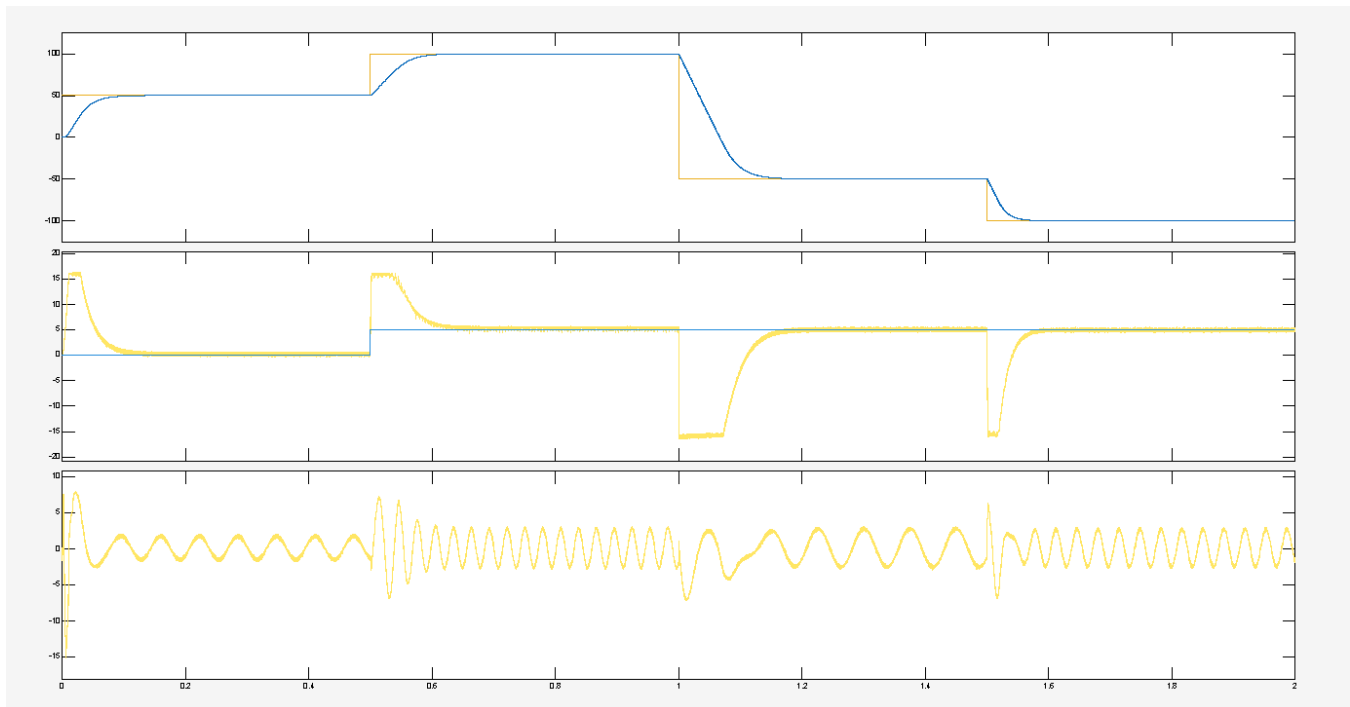


Figure 1.5: Simulation results showing the dynamic response of the DTC-controlled induction motor. The top plot shows the motor speed tracking changes in the reference speed, while the middle plot shows the electromagnetic torque adjusting quickly to match the new operating conditions. The bottom plot shows the motor current waveform, which becomes more active during speed changes as the controller works to maintain the desired response.

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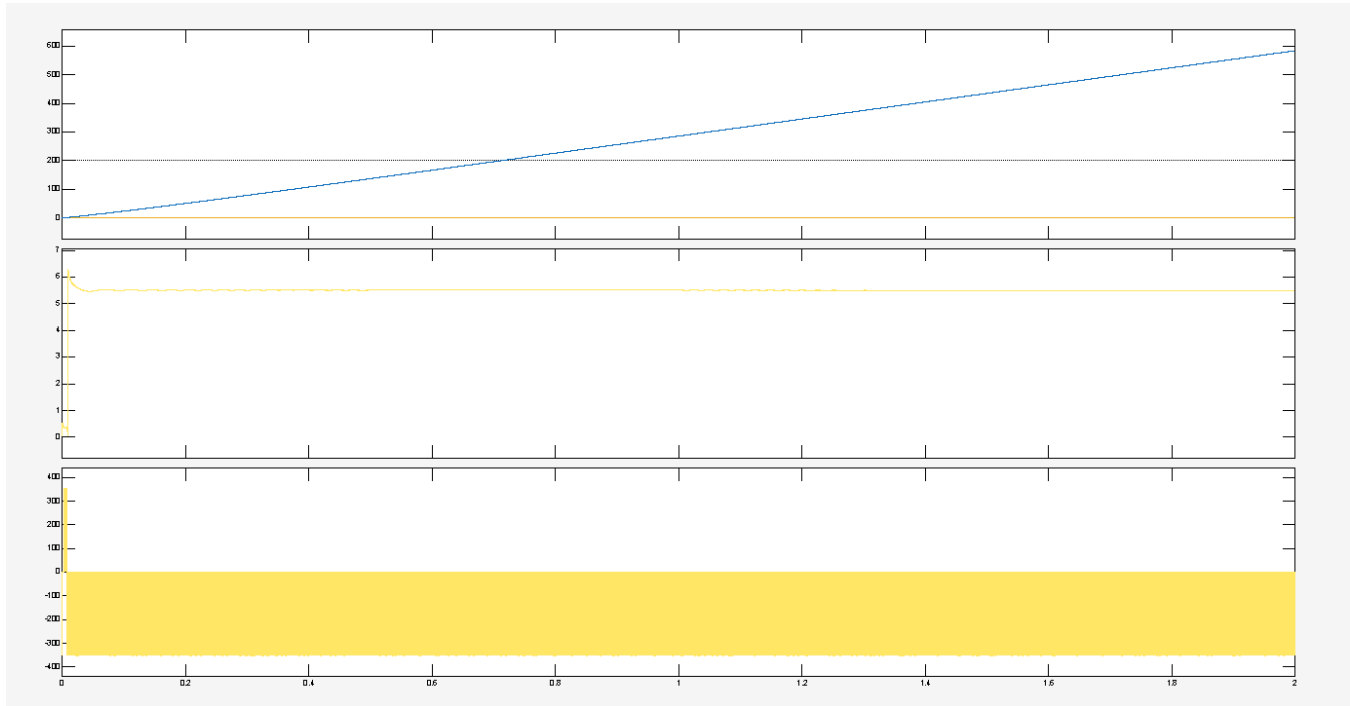


Figure 1.6: Simulation results showing stator flux behavior in the DTC system. The top plot shows the magnitude of the stator flux increasing over time, while the middle plot shows the flux angle stabilizing after a short transient. The bottom plot represents the sector or switching state, indicating how the controller selects switching regions to maintain the desired flux.

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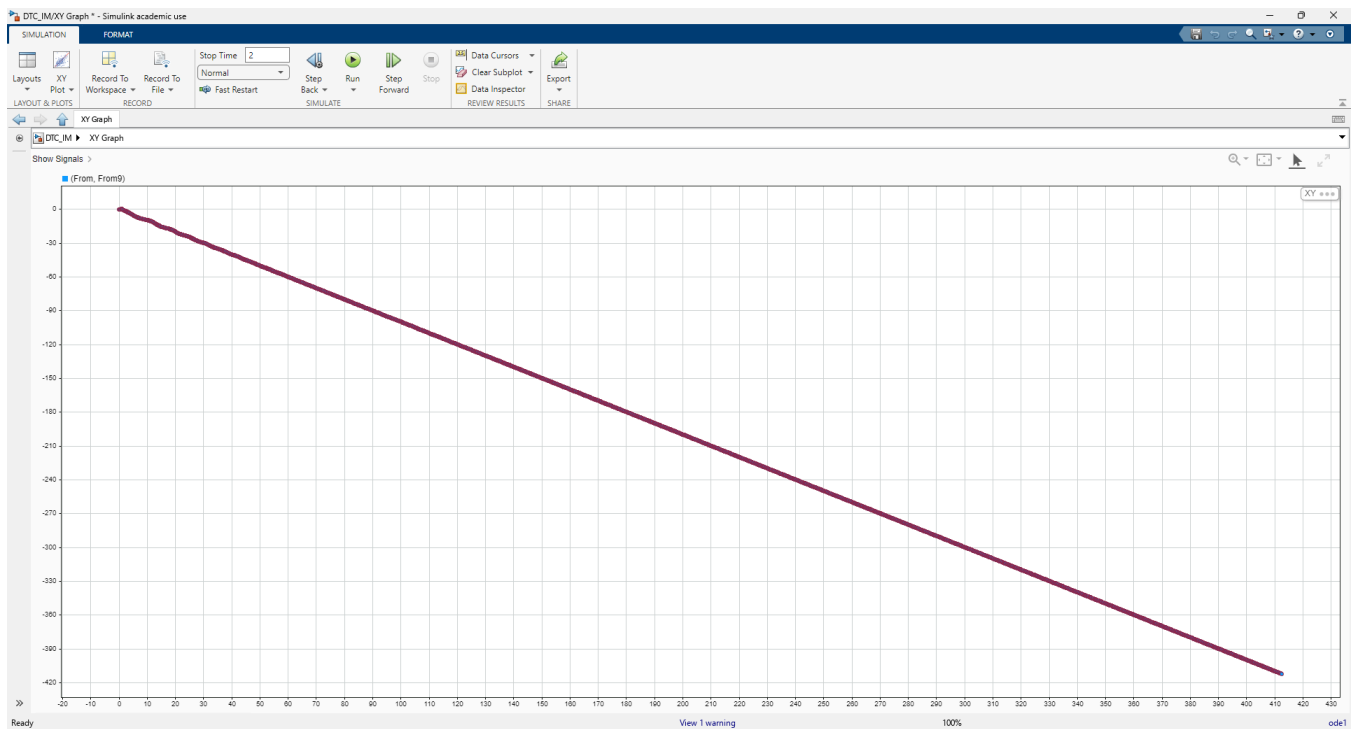


Figure 1.7: XY plot of the stator flux components in the $\alpha\beta$ reference frame. The plot shows the trajectory of the flux vector over time, which should ideally form a circular pattern for constant flux magnitude. The linear trend observed indicates a change in operating conditions or flux magnitude during the simulation.